

Multilingual DistilWhisper:
Efficient Distillation of Multi-task Speech Models via
Language-Specific Experts

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1. Abstract

Whisper [1] is a multitask and multilingual speech model covering 99 languages. It yields commendable ASR results in a subset of its covered languages, but the model still underperforms on a non-negligible number of under-represented languages, a problem exacerbated in smaller model versions. In this work, we propose **Multilingual DistilWhisper**, an approach able to bridge the performance gap in ASR for these languages while retaining the advantages of multitask and multilingual capabilities.

Our approach involves two key strategies: lightweight modular ASR fine-tuning of *whisper-small* using language-specific experts, and knowledge distillation from *whisper-large-v2*. This dual approach allows us to **effectively boost ASR performance while keeping the robustness** inherited from the multitask and multilingual pre-training.

Results demonstrate that our approach is more effective than standard fine-tuning or LoRA adapters, boosting performance in the targeted languages for both in- and out-of-domain test sets, **while introducing only a negligible parameter overhead at inference**.

2. Multilingual DistilWhisper Approach.

We propose the **Multilingual DistilWhisper** approach: we plug **conditional language-specific routing (CLSR)** modules [2] into a small Whisper (*whisper-small*), and **optimize these modules jointly on ASR fine-tuning and knowledge distillation (KD)** from a larger Whisper (*whisper-large-v2*).

WHY A DUAL APPROACH?

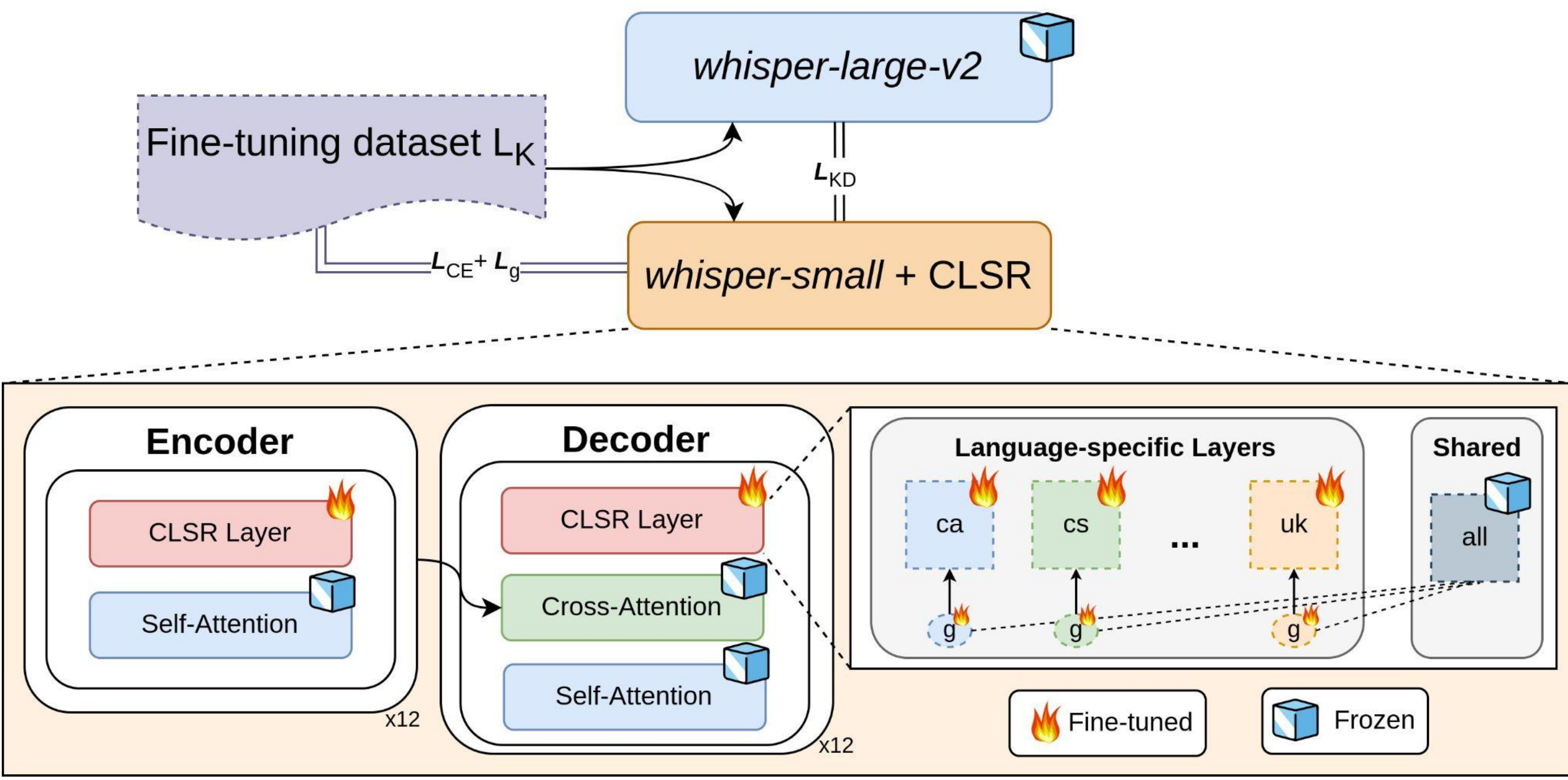
CLSR modules combine existing Whisper modules with extra language-specific parameters

- leverage robustness acquired by whisper pre-training
- benefit from extra parameters to further specialize for underperforming languages

Knowledge Distillation further improves robustness and reduces dataset bias during CLSR tuning.

HOW DO WE MODEL THIS?

- **Lightweight training:** We add **language-specific gates and layers** at the feed-forward layers. Only CLSR is trained, all the remainder is kept frozen. Language specific gates allow us to train different languages in **parallel** in a single V100 32GB. See Figure below:



3. Datasets

- **Fine-tuning data (in-domain):** We downsample the **CommonVoice 13.0** (CV-13) dataset [3] for target languages (10k utterances; 14h hours).
- **Out-of-domain test set:** We use the **FLEURS** dataset [4] for both a good language overlap with CV-13, and an effective out-of-domain ASR setting.
- **Language selection** was performed based on availability on both datasets and amount of data: **Catalan (ca), Czech (cs), Galician (gl), Hungarian (hu), Polish (pl), Thai (th), Tamil (ta) and Ukrainian (uk)**.

4. Results

Baselines: *whisper-large-v2* (*KD teacher, 4x larger*), and *whisper-small*;
Adaptation approaches: the parameter-light LoRA approach [5] (**LoRA-FT**) and full fine-tuning (**FT**);
Our method: CLSR fine-tuning (**CLSR-FT**) and the full proposed approach (**DistilWhisper**).

Table: Average WER (↓) for 8 languages (ca, cs, gl ,hu, pl, ta, th, uk). Adaptation methods (middle and bottom) train on 14h of speech for CV-13 (in-domain), and are evaluated on FLEURS (out-of-domain).

model	# params	FLEURS avg	CV-13 avg
<i>whisper-large-v2</i>	1.5B	12.5	14.9
<i>whisper-small</i>	244M	28.3	31.4
<i>whisper-small</i> + FT	244M	23.3 ±0.06	16.3 ±0.09
<i>whisper-small</i> + LoRA-FT	379M	24.9 ±0.07	18.2 ±0.02
<i>whisper-small</i> + CLSR-FT	369M	23.4 ±0.19	16.3 ±0.08
DistilWhisper	369M	22.8 ±0.21	16.0 ±0.04

DistilWhisper outperforms other adaptation approaches.

- **Our proposed approach outperforms** both LoRA adaptation and full fine-tuning the whisper model for in and out-of-domain test sets.

Knowledge distillation results in better out-of-domain generalization.

- Compared to CLSR-FT, knowledge distillation further improves results, helping particularly for out-of-domain generalization (avg -0.6).

Reducing the performance gap for models with limited capacity.

- We **reduce the out-of-domain performance gap** between *whisper-large-v2* and *whisper-small* by **35.2%** (avg -5.5) with a parameter overhead at inference time of only **10%** (25 M).

A very data light approach!

- Check out our paper experiments to see that **our approach can be effective with as little as 4h of speech as training data!**

5. Use Multilingual DistilWhisper today!

Download our language experts, code and paper:


CODE


WEIGHTS


PAPER

It is very easy to train new language experts, and use existing ones! Check the code and weights above.

References

[1] Alec Radford, et al. “Robust speech recognition via large-scale weak supervision” in International Conference on Machine Learning. PMLR, 2023.
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[4] Alexis Conneau, et al. “Fleurs: Few-shot learning evaluation of universal representations of speech” in IEEE SLT 2022.
[5] Edward J Hu, et al., “LoRA: Low-rank adaptation of large language models” in International Conference on Learning Representations, ICLR, 2021.

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